

Project: JusticeServe - Legal Case & Court

Management System

1. Introduction

JusticeServe is a web-based platform designed for judiciary bodies, courts, and government

agencies to manage legal cases, hearings, and compliance. It enables citizens to file cases,

lawyers to manage proceedings, judges to oversee hearings, and administrators to monitor court operations.

The system supports workflows for citizen registration, case filing, hearing scheduling, judgment recording, compliance monitoring, and analytics. It ensures transparency and accountability by maintaining audit trails, dashboards, and performance metrics across judicial governance.

Actors / Users

- Citizen: Files cases, tracks case progress, receives updates.
- Lawyer: Manages cases, submits documents, represents clients.
- Judge: Oversees hearings, delivers judgments, monitors compliance.
- Court Clerk: Manages scheduling, documentation, and case records.
- Administrator: Configures workflows, manages users, oversees reporting.
- Compliance Officer: Ensures adherence to judicial policies, audits records.
- Government Auditor: Reviews compliance reports, monitors court efficiency.

2. Module Overview

2.1 Identity & Access Management

2.2 Citizen Registration & Profile Management

2.3 Case Filing & Document Management

2.4 Hearing Scheduling & Court Proceedings

2.5 Judgment & Order Management

2.6 Compliance & Audit Management

2.7 Reporting & Analytics

2.8 Notifications & Alerts

3. Architecture Overview

- Frontend: Angular or React for responsive dashboards.
- Backend: REST API-based microservices (Identity, Citizens, Cases, Hearings, Judgments, Compliance, Reports).
- Database: Relational DB (MySQL/PostgreSQL/SQL Server).
- Deployment: Cloud/on-prem with API gateway, WAF, centralized logging.

4. Module Wise Design

4.1 Identity & Access Management

Features

- Secure authentication and role-based access control.
- Audit logging of all actions.

Entities

- User(UserID, Name, Role [Citizen/Lawyer/Judge/Clerk/Admin/Compliance/Auditor], Email, Phone, Status)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp)

4.2 Citizen Registration & Profile Management

Features

- Registers citizens and maintains profiles.
- Validates identity and legal documents.

Entities

- Citizen(CitizenID, Name, DOB, Gender, Address, ContactInfo, Status)

- CitizenDocument(DocumentID, CitizenID, DocType [IDProof/LegalDoc], FileURI, UploadedDate, VerificationStatus)

4.3 Case Filing & Document Management

Features

- Citizens and lawyers file cases.
- Court clerks validate and manage case records.
- Track case status and associated documents.

Entities

- Case(CaseID, CitizenID, LawyerID, Title, Description, FiledDate, Status)
- CaseDocument(DocumentID, CaseID, DocType [Petition/Evidence], FileURI, UploadedDate, VerificationStatus)

4.4 Hearing Scheduling & Court Proceedings

Features

- Schedule hearings for cases.
- Judges and clerks manage proceedings.
- Track hearing outcomes.

Entities

- Hearing(HearingID, CaseID, JudgeID, Date, Time, Status)
- Proceeding(ProceedingID, HearingID, Notes, Date, Status)

4.5 Judgment & Order Management

Features

- Judges deliver judgments and issue orders.
- Track compliance with court orders.

Entities

- Judgment(JudgmentID, CaseID, JudgeID, Summary, Date, Status)
- CourtOrder(OrderID, CaseID, JudgeID, Description, Date, Status)

4.6 Compliance & Audit Management

Features

- Ensure adherence to judicial policies.
- Maintain compliance records.
- Conduct audits with findings.

Entities

- ComplianceRecord(ComplianceID, EntityID, Type [Case/Hearing/Judgment], Result, Date, Notes)
- Audit(AuditID, OfficerID, Scope, Findings, Date, Status)

4.7 Reporting & Analytics

Features

- Generate dashboards for cases, hearings, judgments, and compliance.
- Provide analytics on judicial efficiency.

Entities

- Report(ReportID, Scope [Case/Hearing/Judgment], Metrics, GeneratedDate)

4.8 Notifications & Alerts

Features

- Send alerts for hearing schedules, judgment updates, compliance deadlines.
- Notify citizens and lawyers about case progress.

Entities

- Notification(NotificationID, UserID, EntityID, Message, Category [Case/Hearing/Judgment/Compliance], Status, CreatedDate)

5. Deployment Strategy

- Local: Angular/React frontend, Spring Boot/.NET Core backend, local DB.

- Production: Cloud/on-prem deployment, API gateway, WAF, centralized logging.

6. Database Design

Tables & Attributes

- User(UserID, Name, Role, Email, Phone, Status)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp)
- Citizen(CitizenID, Name, DOB, Gender, Address, ContactInfo, Status)
- CitizenDocument(DocumentID, CitizenID, DocType, FileURI, UploadedDate, VerificationStatus)
- Case(CaseID, CitizenID, LawyerID, Title, Description, FiledDate, Status)
- CaseDocument(DocumentID, CaseID, DocType, FileURI, UploadedDate, VerificationStatus)
- Hearing(HearingID, CaseID, JudgeID, Date, Time, Status)
- Proceeding(ProceedingID, HearingID, Notes, Date, Status)
- Judgment(JudgmentID, CaseID, JudgeID, Summary, Date, Status)
- CourtOrder(OrderID, CaseID, JudgeID, Description, Date, Status)
- ComplianceRecord(ComplianceID, EntityID, Type, Result, Date, Notes)
- Audit(AuditID, OfficerID, Scope, Findings, Date, Status)
- Report(ReportID, Scope, Metrics, GeneratedDate)
- Notification(NotificationID, UserID, EntityID, Message, Category, Status, CreatedDate)

7. User Interface Design

- Citizen Portal: Case filing, document submission, case tracking.
- Lawyer Dashboard: Case management, document upload, hearing updates.
- Judge Console: Hearing management, judgment delivery, compliance tracking.
- Court Clerk Panel: Scheduling, documentation, case record management.
- Administrator Panel: Workflow setup, compliance monitoring, reporting.

- Compliance Officer Console: Policy monitoring, audits, compliance reporting.
- Government Auditor Dashboard: Judicial program review, compliance monitoring.

8. Non-Functional Requirements

- Performance: Handle 150,000 concurrent users across judicial networks.
- Security: Role-based access, encrypted data storage, immutable audit logs.
- Scalability: Support nationwide rollout across multiple courts.
- Availability: 99.9% uptime.
- Maintainability: Modular microservices, API versioning, automated migrations.
- Observability: Centralized logging, KPIs (case resolution time, hearing efficiency, compliance adherence).

9. Assumptions & Constraints

- Initial rollout for a single court system before nationwide expansion.
- Notifications limited to in-app and SMS/email alerts.
- Fully implementable using Java Spring Boot/ASP.NET Core, Angular/React, and relational DB